Prompt Position, Not Prompt Content: Three Ordering Failures in a Thai Retrieval-Augmented Question Answering System

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***Abstract*—Evaluations of retrieval-augmented generation (RAG) routinely attribute answer quality to the retriever. We report three cases from a production-intent Thai RAG system in which the number being measured was set by where an instruction sat in the prompt rather than by retrieval, and in which the prompt's content was held fixed while only its ordering changed. On a corpus of 2,853 Thai academic-council resolutions with 106 entity-anchored queries and 1,046 relevance judgments (9.87 relevant documents per query), we find: (i) the conventional fix of placing instructions *after* the retrieved context interacts with the serving runtime's front-truncation policy to convert a loud failure into a silent one, damaging 81 of 1,590 generated answers with no observable symptom; (ii) a brevity instruction ("answer in at most three sentences") acts as a citation-recall cap, and replacing it raises citation recall significantly on two of four answering arms (up to $+0.0871$, Holm-adjusted $p = 0.0000$, family size $m = 9$) with no change to retrieval; and (iii) a prohibition already present in the prompt is defeated by a later, more recent rule — moving it after the offending rule and declaring its precedence takes closed-book hallucinations from 24 of 106 to 1 of 106 on `gemma4:e4b`, and phantom citations from 37/37 to 1/1, while leaving the four answering arms essentially unchanged. All three effects are invisible to retrieval metrics. We give the detection signature for (i), report every comparison with its Holm family size, and state the generator noise floor that bounds what a single-cell difference can mean.**

***Index Terms*—retrieval-augmented generation, prompt engineering, Thai NLP, information retrieval evaluation, citation grounding, hallucination**

I. Introduction

A retrieval-augmented question answering system has two halves that are usually evaluated separately: a retriever, scored with ranking metrics against relevance judgments, and a generator, scored on the answer it produces from the retrieved context [1]. When end-to-end answer quality is reported, the discussion almost always concerns the retriever — which embedder, which fusion, which reranker.

This paper reports three findings from building a Thai RAG system over institutional meeting minutes, in each of which the *prompt's layout*, not its content and not the retriever, determined the measured outcome. The three are not variations of one effect; they act at different layers, and each would have been reported as a different kind of result had it not been isolated:

1) **Layout versus runtime truncation.** Placing instructions after the retrieved documents is the standard defence against an over-long prompt losing its rules. On our serving runtime it makes truncation *invisible instead of harmless*: the rules always survive, the best-ranked evidence is deleted, and the answer is well-formed, correctly shaped in its citations, and evidence-poor for reasons nothing in the output reveals.
2) **A brevity instruction is a recall cap.** Against a judgment set averaging 9.87 relevant documents per query, an instruction to answer in at most three sentences cannot cite them. The resulting flat citation recall reads as a generator ceiling and is a prompt budget.
3) **A rule's position beats its presence.** A rule forbidding answers that the supplied documents do not support was identical, word for word, in both prompt variants we compare. Under one variant a second model fabricated answers on 24 of 106 zero-document queries. The repair was not a new prohibition but the same prohibition moved after the rule that was defeating it, with an explicit precedence declaration.

Effect (1) is a systems interaction; effect (2) is a measurement artifact that would silently understate any RAG system evaluated this way; effect (3) is a safety failure with a one-line fix. What unites them is that in all three the prompt's *content* was adequate and its *order* was not.

Contributions. (i) Three measured prompt-ordering effects in a non-English, entity-anchored RAG setting, each isolated with the rest of the system held fixed; (ii) an exact, cheap detection signature for silent prompt truncation under a widely used local-inference runtime; (iii) a prompt-repair rule — *declare precedence and move the rule after its competitor, rather than adding a stronger prohibition* — validated on two generators with a built-in control; and (iv) an explicit account of the generator noise floor and of multiple-comparison family sizes, so that the null cells reported here can be read as bounds rather than as absence of effect.

II. Related Work

Thai retrieval resources. Thai is covered as a *language* by several multilingual retrieval benchmarks. MIRACL [2] includes Thai Wikipedia ad-hoc retrieval with natural queries

TABLE I
EVALUATION QUERY SET

Query type	Queries
`course`	33
`person`	30
`program`	30
`faculty` (aggregate)	13
Total	106
Relevance judgments	1,046
Mean relevant per query	9.87

written by native speakers, at roughly 9–10 labels per query; Mr. TyDi [3] includes Thai but is shallow, at about one positive passage per query. BEIR [4] is English-only. Domain-specific Thai RAG evaluation exists: NitiBench [5] benchmarks LLM frameworks on Thai legal question answering. What is not covered is the *domain* we work in — administrative and academic-governance minutes — and, more importantly for this paper, none of these resources is designed to separate prompt effects from retrieval effects, because none of them evaluates a generator at all.

Position effects in language models. That transformer language models use long contexts unevenly is established: Liu et al. [6] show a U-shaped curve in which information at the beginning and end of a context is used far more effectively than information in the middle. Our findings are adjacent but distinct in two ways. First, the object whose position matters here is an *instruction*, not evidence: in finding (3) two rules compete and the later one wins even though the earlier one is more specific and strictly prohibitive. Second, finding (1) is not a model property at all — it is an interaction between prompt layout and a serving runtime's truncation policy, and it therefore survives any improvement in the model's use of long context.

Retrieval components. Our arms use BM25 [7], dense retrieval with multilingual embedders including BGE-M3 [8], and reciprocal rank fusion [9] for the hybrid arms. These are background: no claim in this paper is about which of them is better.

III. Experimental Setup

A. Corpus and query set

The corpus is 2,853 Thai academic-council resolutions, obtained as scanned PDFs and converted by OCR. Relevance is judged at the level of a *resolution* — the unit a reader of these minutes actually wants — rather than at the level of a retrieval chunk.

The evaluation set contains 106 entity-anchored queries with 1,046 relevance judgments, i.e. a mean of **9.87 relevant documents per query**. This depth is the reason effect (2) is visible at all: against a shallow judgment set of about one relevant document per query, an instruction capping the answer at three sentences costs nothing measurable. Composition is given in Table I.

B. Arms

Five arms supply the context. Four retrieve ten documents each: `hybrid` (RRF over BM25 and a dense arm), `dense`, `bm25`, and a deliberately weak `m2v` arm using a low-capacity static embedder. The fifth, `closed_book`, supplies *no documents at all* and is the control that carries finding (3): with an empty context, the only correct behaviour is to abstain, and any answer produced is by construction ungrounded.

C. Generators

Both generators run locally through the same serving runtime (ollama) on a single consumer GPU. `phi4` is the primary generator; `gemma4:e4b` is a second-generator check, run precisely so that a prompt effect can be told apart from a model idiosyncrasy. Sampling temperature is 0. Following a measurement reported in Section VII, temperature 0 is *not* treated here as reproducible.

D. Metrics and statistics

Citation recall is the fraction of a query's gold resolutions that the answer's citation line names, macro-averaged over all 106 queries. *Citation precision* is the fraction of cited documents that are gold, macro-averaged over queries that cited at least one document. A *phantom* citation is a label that does not appear in the supplied context at all, reported as a micro count over all citations. For finding (3) we also report the abstention 2×2: whether the context contained gold, against whether the model abstained; the cell with no gold and no abstention is a hallucination.

Significance is paired bootstrap over queries (10,000 resamples, fixed seed) with Holm correction [10]. **We report the Holm family size m with every adjusted p-value**, because in this study the identical test on identical data reads significant in one family and not in another purely through m; a bare adjusted p is ambiguous.

IV. Finding 1: Layout Interacts With Runtime Truncation

A. The loud failure, and the standard fix

An initial pilot placed the instructions *before* the retrieved documents. On prompts that carried context it produced **0 of 4** correctly formatted citations, while producing **4 of 4** on the short zero-document prompts — same model, same instructions, so not a capability limit. The cause is that the runtime truncates an over-long prompt from the *front*: with instructions first, the rules were cut away before the model ever saw them. The standard fix applies — put the documents first and the rules last, so a truncation cannot delete the rules.

B. What the fix actually did

The fix worked, and it converted a loud failure into a silent one. Because the prompt lays the retrieved documents out best-ranked first and the rules last, an over-long prompt now loses *the highest-ranked evidence* and keeps every rule. The result is an answer that is well-formed, cites in the correct format, and abstains when told to — and that is evidence-poor for reasons

TABLE II
PROMPT TRUNCATION, ONE 14,721-TOKEN PROMPT

num_ctx	prompt_eval_count	interpretation
4,096	2,050	cut to num_ctx$/2 + 2$
8,192	4,098	cut to num_ctx$/2 + 2$
16,384	14,721	fed whole

nothing in the output reveals. Nothing warned; the runtime reports no error.

C. The truncation rule, measured

We measured the runtime's behaviour rather than reading it from documentation. One 14,721-token prompt at three context sizes gives Table II.

An intuitive hypothesis — *the runtime never feeds more than half the context window* — is refuted by its own control: prompts of 5,651, 6,885 and 7,508 tokens are fed *whole* at num_ctx $=$ 8,192. The rule that reproduces all six observations is:

> A prompt that fits num_ctx is fed whole; a prompt that exceeds it is cut to num_ctx$/2 + 2$ tokens, keeping the *tail*.

The practical value of the second clause is that prompt_eval_count $=$ num_ctx$/2 + 2$ is an **exact detection signature**. It costs nothing: the runtime already returns prompt_eval_count with every response. We therefore recommend recording it per generated answer and asserting the signature, in preference to any layout convention — a layout rule chooses *which half* of the prompt dies, and cannot tell you *that* something died.

D. Blast radius

Applying the signature retrospectively identified **81 of 1,590** generated answers as produced from a truncated prompt, with as little as 31% of the intended prompt actually fed. Regenerating all 81 at num_ctx $=$ 16,384 flipped **1 of 57** verdicts in the primary report and **3 of 57** in a secondary one; the prompt-ablation headline of Section V did not move. We stress the direction of that result: the damage was real and the published conclusions largely survived it, which is exactly the case in which a silent fault is never found by inspecting outcomes.

V. FINDING 2: A BREVITY INSTRUCTION CAPS CITATION RECALL

A. Design

The prompt is a numbered rule list. We hold rules 1–3 and the answer format fixed and change only rule 4, giving the variants in Table III. Rules 1–3 state, in Thai: (1) answer only from the supplied documents, never from other knowledge; (2) cite the documents used by bracketed number; (3) if the supplied documents do not contain the answer, reply NOT-FOUND only, and do not guess.

TABLE III
THE THREE PROMPT VARIANTS (ENGLISH GLOSS; ORIGINALS ARE THAI)

sentence_cap — the baseline (shown as cap in later tables)
4. Answer briefly, in no more than 3 sentences.
cite_all
4. Cite every relevant document found in the supplied documents, not just one; the answer length is unlimited so long as it covers every relevant one.
cite_all_guarded — adds rules 5 and 6 (shown as guarded in later tables)
4. *Identical to* cite_all. 5. If no supporting documents were supplied at all, reply NOT-FOUND and give the citation as "–" only; cite no number whatsoever. **This rule outranks rule 4.** 6. Cite only numbers that actually appear in the documents above; never invent a number.

TABLE IV
CITATION PRECISION / RECALL, PHI4, 106 QUERIES. VARIANT NAMES ABBREVIATED AS IN TABLE III

Arm	Variant	Precision	Recall
hybrid	cap	0.7013	0.2952
hybrid	cite_all	0.7185	**0.3823**
hybrid	guarded	0.6945	0.3794
dense	cap	0.6867	0.2738
dense	cite_all	0.6381	0.3145
dense	guarded	0.6650	**0.3481**
bm25	cap	0.6591	0.2329
bm25	cite_all	0.5801	**0.3118**
bm25	guarded	0.5968	0.2733
m2v	cap	0.5417	0.1807
m2v	cite_all	0.5057	**0.1989**
m2v	guarded	0.4539	0.1761

B. Result

Table IV gives citation precision and recall for phi4 across all four answering arms and all three variants, and Table V the significance of the baseline-to-cite_all contrast.

Citation recall rises significantly on hybrid ($+0.0871$) and bm25 ($+0.0789$), both at Holm-adjusted $p = 0.0000$ with $m = 9$. **No retrieval changed**: the contexts are byte-identical between variants and only rule 4 differs. The pre-existing reading of the flat baseline recall — that the generator could not do better — is therefore wrong. It was a budget imposed by the prompt.

C. Two qualifications we do not want cited away

First, the dense arm's gain is **not significant** ($+0.0407$, Holm $p = 0.6610$). We report it as a bound, not as an effect: the data rule out a loss and do not establish a gain. The mechanism we suspect — that a stronger context reduces the marginal value of the instruction — is consistent with dense and hybrid being the two strongest arms, but we did not test it and do not claim it.

TABLE V

SENTENCE_CAP → CITE_ALL, PHI4. PAIRED BOOTSTRAP, HOLM-CORRECTED, FAMILY SIZE $m = 9$

Arm	Metric	Diff	95% CI	Holm p
hybrid	recall	$+0.0871$	$[+0.0438, +0.1307]$	**0.0000**
bm25	recall	$+0.0789$	$[+0.0440, +0.1174]$	**0.0000**
dense	recall	$+0.0407$	$[-0.0133, +0.0932]$	0.6610
m2v	recall	$+0.0182$	$[-0.0136, +0.0489]$	0.7398
bm25	precision	-0.0644	$[-0.1155, -0.0134]$	0.0798
hybrid	precision	$+0.0043$	$[-0.0559, +0.0636]$	1.0000

TABLE VI

CLOSED_BOOK ARM: EMPTY CONTEXT, SO ABSTENTION IS THE ONLY CORRECT ANSWER. VARIANTS ABBREVIATED AS IN TABLE III

Model	Variant	Halluc.	Abstained	Phantom
`phi4`	`cap`	0	106	0/0
`phi4`	`cite_all`	2	104	5/5
`phi4`	`guarded`	**0**	106	0/0
`gemma4:e4b`	`cite_all`	24	82	37/37
`gemma4:e4b`	`guarded`	**1**	105	1/1

Second, the gain is a *trade*. Citation precision falls on three of four arms, most on `bm25` (-0.0644), and while no precision cell is significant at $m = 9$, the direction is consistent. Instructing a model to cite everything relevant produces more citations, and some of the extra ones are wrong.

D. The effect is not model-specific, but the levels are

Under `cite_all`, `gemma4:e4b` reaches citation recall 0.4956 (hybrid) and 0.5049 (dense), against `phi4`'s 0.3823 and 0.3145. The direction of the prompt effect transfers between generators; **the levels do not transfer at all**. Any absolute citation-recall figure reported for a RAG system is a joint property of the retriever, the prompt and the generator, and is not portable across a change in any of the three.

VI. Finding 3: A Rule Already Present, Defeated by Position

A. The failure

Rule 3 — *if the supplied documents do not contain the answer, reply* NOT-FOUND *only, and do not guess* — is present, identical and word for word, in all three variants. It is a strict prohibition and it is stated before any competing instruction.

It does not hold. Table VI gives the `closed_book` arm, where the context is empty and abstention is the only correct output. Under `cite_all`, `gemma4:e4b` answers anyway on **24 of 106** queries, and *every one* of the 37 citations it produces is a phantom — a bracketed document number when no document was supplied at all. `phi4` shows the same failure at much lower magnitude: 2 of 106, 5/5 phantom.

B. The diagnosis: recency, not absence

The natural reading is that the model ignores instructions. The comparison refutes it. Rule 3 is *identical* between `sentence_cap` and `cite_all`, and under `sentence_cap` `phi4` hallucinates 0 times. What changed is the rule that follows it: `cite_all`'s rule 4 instructs the model to cite every relevant document, it is the last line before the question, and *with nothing to cite it reads as an instruction to produce citations*. The specific, prohibitive, earlier rule loses to the general, permissive, later one.

C. The repair, and why its shape matters

The repair is not a stronger prohibition. It is the same prohibition, moved: rule 5 restates the zero-document case *after* rule 4 and declares precedence in as many words — *this rule outranks rule 4*. Rule 6 adds a ceiling on which labels are citable, aimed at the bounded-context version of the same over-production (the `dense` arm produced 4 phantom labels against 5 supplied documents).

The effect is in Table VI: **24 hallucinations become 1**, and 37/37 phantom citations become 1/1. `phi4` goes from 2 to 0.

D. The control that makes this readable

A guard that suppressed hallucination by making the model reticent everywhere would be worthless. The four answering arms are the built-in control, and they barely move: for `gemma4:e4b` the abstention 2×2 is *identical* between `cite_all` and `cite_all_guarded` on `hybrid` (93/9/2/2) and on `dense` (92/8/4/2), and moves by at most three cells on `bm25` and `m2v`. Citation recall on the strong arms is unchanged or slightly higher under the guard (hybrid $0.4956 \rightarrow 0.5260$). The guard reached the case it was aimed at and left the rest alone.

E. Where the failure concentrates

In *both* models, 100% of closed-book hallucination occurs on `course` queries. We report this as an observation, not a mechanism: a course code is a short, highly patterned string about which a language model has strong priors, and the plausible explanation — that a well-formed guess is easier to produce for this query shape — was not tested here.

VII. Limitations and Threats to Validity

Temperature 0 is not reproducible, and we measured how badly. Feeding byte-identical prompts to `phi4` at temperature 0 reproduces the identical citation set only **14 of 24** times under `cite_all`. Every claim in this paper is therefore made over families of 106 queries with Holm-corrected paired bootstrap, and no single-cell difference is interpreted. A reader reproducing this work should expect cell-level disagreement, and should re-measure this noise floor rather than infer determinism from the temperature setting.

Holm family size is load-bearing. In our data the identical test on identical data reads significant in a family of 9 and not significant in a family of 24. We quote m everywhere; a comparison from Table V should not be quoted against a p-value computed in a different family.

A generation cap can masquerade as a retrieval result. The maximum generated length is capped at 4,096 tokens. **3 of 1,272** `phi4` cells reach that cap, all of them `course` queries. A capped answer never reaches its citation line and is therefore scored as having cited nothing — a generator failure that reads as a retrieval failure. The count is small enough not to move any conclusion here, and is stated because the failure mode is invisible in the metric.

The judgments are rule-derived. Relevance judgments were derived by string containment of each query's anchor entity, which under-counts documents phrasing that entity differently. Citation recall inherits the undercount. It is a level effect shared by every arm and variant, and the comparisons here are paired, so it does not threaten the contrasts — but absolute values should be read as a lower bound.

Scope. One corpus, one language, one serving runtime, two generators. Finding (1) is specific to a runtime that truncates from the front and should be re-measured, not assumed, elsewhere; the *method* — assert a signature on the runtime's own reported token count — transfers. Findings (2) and (3) were each observed on two generators, with the direction preserved and the levels not.

VIII. Conclusion

Three times in one system, the quantity we were measuring was set by where an instruction sat rather than by what it said. Placing rules after the context is the correct fix for one failure and silently creates another, and the durable defence is a truncation signature rather than a layout convention. A brevity instruction, harmless against a shallow judgment set, acts as a citation-recall cap against a deep one — and the ceiling it produces is easily mistaken for a generator limit. And a prohibition already written into the prompt is defeated by a later rule, with a repair that consists of moving it and declaring precedence rather than strengthening it.

For practitioners the operational form is short: record the runtime's reported prompt token count and assert it; check whether an instruction constrains the quantity you are about to report; and when a rule is disobeyed, look at what follows it before rewriting it. For evaluation the point is sharper — a RAG result that does not hold the prompt fixed, and say so, may be reporting a property of its instructions.

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